



Yasuda Koichiro <k.yasuda5160@gmail.com>

you are conceptually confused-

11 件のメッセージ

Graciela Gelmini <gelmini@physics.ucla.edu>

2026年9月19日 14:12

To: Danny Marfatia <dmarf8@hawaii.edu>, Muping Chen <mpchen1864@gmail.com>, Yasuda Koichiro <k.yasuda5160@gmail.com>

Danny:

we attempted two different methods:

---Method 1: fit Φ up to energies above 2 MeV: we had a persistent numerical problem at high energies, one that we had encountered at low energies too. Instead of solving this remaining issue we found that method 2 did not require solving it.

--- Method 2: fit $\Delta\Phi$ up to 2 MeV. This is a self consistent method **and we wrote the paper about method 2.**

You want to go back to method 1 (not presented in the paper) and say that we were too lazy to solve the numerical problem at high energies.

We did not include Method 1 in the paper. This is why your sentence is **wrong**:

you are confusing both methods: so you said we fitted **$\Delta\Phi$ at $E > 2$ MeV** where is is **not defined-** (or negative if you extend the definition)

We wrote a paper about method 2, self consistent.

If you want to return to Method 1, introduce it in the paper and solve all its problems- this is not the paper we wrote.

When talking about the bias/variance in method 2 I talked about the error in method 2.

YOUR SENTENCE ABOUT METHOD 1 DOES NOT MAKE SENSE WITHIN METHOD 2.

Best,
Graciela

Yasuda Koichiro <k.yasuda5160@gmail.com>

2026年9月19日 16:41

To: Graciela Gelmini <gelmini@physics.ucla.edu>

Cc: Danny Marfatia <dmarf8@hawaii.edu>, Muping Chen <mpchen1864@gmail.com>

Dear Danny, Graciela,

I confirm Danny's recollection. In the early setup we used to fit $\delta\Phi$ with intervals extending above 2 MeV, where $\delta\Phi$ should be zero by definition, since the rate from neutrinos above 2 MeV is already in h_i (a nonzero value there would double count it). The grid was uniform only below 2 MeV. The minimization did not return zero above 2 MeV stably, namely the step structure there was unstable.

At the same time, the method the paper currently describes is self-consistent as written. The signal is $\delta\Phi$ below 2 MeV, and h_i is exactly the rate from neutrinos above 2 MeV.

Best,
Ko

2026年9月19日(土) 14:12 Graciela Gelmini <gelmini@physics.ucla.edu>:
[元のメッセージ非表示]

Danny Marfatia <dmarf8@hawaii.edu>
To: Yasuda Koichiro <k.yasuda5160@gmail.com>
Cc: Graciela Gelmini <gelmini@physics.ucla.edu>, Muping Chen <mpchen1864@gmail.com>

2026年9月19日 17:06

Good, now we just have to clearly say what we did.
[元のメッセージ非表示]

Danny Marfatia <dmarf8@hawaii.edu>
To: Yasuda Koichiro <k.yasuda5160@gmail.com>
Cc: Graciela Gelmini <gelmini@physics.ucla.edu>, Muping Chen <mpchen1864@gmail.com>

2026年9月19日 17:49

Ko, wasn't the step structure unstable even below 2 MeV unless we truncated the kernels?
[元のメッセージ非表示]

Yasuda Koichiro <k.yasuda5160@gmail.com>
To: Danny Marfatia <dmarf8@hawaii.edu>
Cc: Graciela Gelmini <gelmini@physics.ucla.edu>, Muping Chen <mpchen1864@gmail.com>

2026年9月19日 17:59

Hi Danny

Yes, they were also unstable.

Best,
Ko
[元のメッセージ非表示]

Danny Marfatia <dmarf8@hawaii.edu>
To: Yasuda Koichiro <k.yasuda5160@gmail.com>
Cc: Graciela Gelmini <gelmini@physics.ucla.edu>, Muping Chen <mpchen1864@gmail.com>

2026年9月20日 10:42

Ko, didn't you take $N_{boot} = 10000$? The text says 500. I recall that the fear of not being taken seriously by experimentalists had prevailed!

[元のメッセージ非表示]

Graciela Gelmini <gelmini@physics.ucla.edu>
To: Yasuda Koichiro <k.yasuda5160@gmail.com>
Cc: Danny Marfatia <dmarf8@hawaii.edu>, Muping Chen <mpchen1864@gmail.com>

2026年9月20日 13:27

Dear Ko,

I just sent a message asking you to please finish the minimization at $E > 2\text{MeV}$. We cannot say that a main motivation for Method 2 is to evade (namely seeing Method 2 as a trick) numerical minimization problems.

There is no reason in the FDS method not to find the steps correctly at high energy.

I had suggested using SMALLER SIZE INTERVALS where the Cr so that all CURLYR IS LARGER SO THE R_{ij} ABOVE 2MeV HAVE THE SAME SIZE AS THOSE BELOW. If the different size of R_{IJ} causes a numerical problem this should solve it.

AI think you never tried this approach, so could you please try? If this does not finx the problem, we are going to find

a way to solve it somehow. It is a numerical problem not and FDS problem.

Method 2 avoids the issue altogether but if Danny insists on mentioning this problem we need to solve it.

The intervals do not need to be of equal size over all energies. Do not add h_i as background. And please do it for no background only.

Best,

Graciela

[元のメッセージ非表示]

Graciela Gelmini <gelmini@physics.ucla.edu>
To: Yasuda Koichiro <k.yasuda5160@gmail.com>
Cc: Danny Marfatia <dmarf8@hawaii.edu>, Muping Chen <mpchen1864@gmail.com>

2026年9月21日 12:21

Dear Ko,

You never tried what I had suggested: changing the size of the intervals in the ansatz so that all R_{ij} are similar. I believe we did not do it because Method 2 (perfectly self consistent- we could have started with it if we would have come up with it from the beginning) makes the issue moot.

I cannot understand why without having tried this approach we are saying that we NEED to eliminate the Curly R at high energy to get good fits.

The only reason I can see for the inclusion of the Higher curly R_i to make problems with minimization is that the R_{ij} are larger. But, let us then take all R_{ij} to be similar. This is an obvious route and I believe will produce good fits. Devoting the paper to Method 2 only we did not need to do it. But if Danny insists on mentioning Method 1, now we need to do it.

See Appendix E. Let us see through this issue.
Best,

Graciela

[元のメッセージ非表示]

Yasuda Koichiro <k.yasuda5160@gmail.com>
To: Graciela Gelmini <gelmini@physics.ucla.edu>
Cc: Danny Marfatia <dmarf8@hawaii.edu>, Muping Chen <mpchen1864@gmail.com>

2026年9月21日 16:20

Dear Graciela, Danny, Muping,

I am performing the Method 1 best fit before we meet, following Graciela's prescription: no h_i background (the flux above 2 MeV is fitted), the no-background setup only, and intervals chosen adaptively above 2 MeV — smaller where the kernels are large — so that the R_{ij} there have the same size as those below 2 MeV.

To make Method 1 exactly consistent with the paper below 2 MeV, I will regenerate the response matrix up to 7 MeV with the same Mathematica pipeline used for the published matrices, on this new grid, and then run the same constrained minimization.

I am available Tomorrow or Wednesday. If you are in the office, can we discuss in person before that?

Best,
Ko

2026年9月21日(月) 12:22 Graciela Gelmini <gelmini@physics.ucla.edu>:
[元のメッセージ非表示]

Graciela Gelmini <gelmini@physics.ucla.edu>
To: Yasuda Koichiro <k.yasuda5160@gmail.com>
Cc: Danny Marfatia <dmarf8@hawaii.edu>, Muping Chen <mpchen1864@gmail.com>

2026年9月21日 18:12

Thanks a lot Ko, Try to work with Muping, if he is available (he has not been answering emails lately). Two people working check each other's mistakes.

It would be fantastic if we could solve the minimization problem for Method 1, not to write another paper, but to show that it is a viable alternative that we did not pursue because Method 2 is simpler and gives us what we wanted (as I said, in principle we did not need to mention Method 1 at all, I believe Method 2 stands on its own).

Graciela

[元のメッセージ非表示]

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Cc: Danny Marfatia <dmarf8@hawaii.edu>, Muping Chen <mpchen1864@gmail.com>

2026年9月21日 20:39

Dear Ko, Thanks again- I wonder if Muping is available to help?

Just two comments:

- we are not going to get the same best fit at $E < E_{\nu}$ as with Method 2, because we have a different minimization procedure, different χ^2 .

The issue is if we get a piecewise constant function for all energies and a number of download steps that is at most $d-1$.

- we need only 1 function (for 1 threshold) just as an example.

Hope it works. Otherwise we may need to understand better how things change with the different sizes of $R_{\{iJ\}}$, why the fit works well in some instances and not in others..

Sorry for the extra work!

Graciela

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